

Stock Analysis Pipeline

Specification Fonctionnelle v1.1

Version:	1.1
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Statut:	Review
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Valideur:	Ced
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Pipeline d'analyse financiere automatisee. Collecte multi-source (yfinance, Finnhub, EDGAR, Seeking Alpha, Tavily, Alpha Vantage), scoring rules-based 40 points (6 categories), generation PDF deep-dive et dossier ZIP auditable. Frontend React + Vite, backend Python FastAPI, expose via Cloudflare Tunnel sur sa.cedlabusa.net.

1. Perimetre et definitions

1.1 Acteurs

ID	Acteur	Description	Competences
ACT-001	Investisseur prin	Utilisateur unique, inve	Analyse financiere, tic
ACT-002	Auditeur externe	Verifie conformite des a	Controle qualite
ACT-003	Veilleur automati	Cron jobs Hermes monitor	Aucune (systeme)

1.2 Perimetre fonctionnel

- IN-001 — Analyse fondamentale par ticker (9 etapes) — Coeur du produit
- IN-002 —

1.3 Hors perimetre

- Trading automatise
- Donnees

2. Exigences fonctionnelles

UC-001 — Analyser un ticker

Acteur: ACT-001 | Preconditions: ticker US valide, API keys, backend actif | Declencheur: saisie ticker + Analyze | Postconditions: PDF + ZIP disponibles

AC-001 (nominal): Given backend actif :8780 and 6 API accessibles, when saisit NVDA et clique Analyze, then pipeline 9 etapes execute, PDF 10-14p genere, ZIP telechargeable.

AC-002 (erreur): Given backend actif, when saisit ZZZZYX, then message erreur < 30s.

AC-003 (partiel): Given API externe down, when analyse lancee, then continue avec sources restantes, signale donnees manquantes.

UC-002 — Consulter le PDF deep-dive

PDF 10-14p: sections ◆ Key Metrics, Financial Health, ● Earnings, Management, ▲ Risks, Sources. Barre entete sombre #2A2A2A, callout boxes bleues #2563EB. Donnee absente = 'DONNEE NON DISPONIBLE'.

UC-003 — Telecharger le dossier ZIP

Bouton Download → ZIP: PDF + analysis.json + sources.json (min 3 fichiers). Audit complet possible.

UC-004 — Batch analysis

POST /api/batch/upload (CSV <= 10 tickers) → POST /api/batch/analyze (sequentiel) → GET status. UI batch en v1.2.

UC-005 — Feedback

POST /api/feedback (message horodate) → GET /api/admin/feedback. Correction scoring via feedback (ADR-002).

3. Modeles de donnees

3.1 Scoring (40 points, rules-based)

Categorie	ID	Pts	Donnees	Logique
Financial Hea	SC-FIN	/10	Dette/equity, current	Ratio → seuil → pts
Growth	SC-GRW	/10	Revenue growth YoY, EP	% croissance → seuil
Valuation	SC-VAL	/8	PER, PEG, P/B, EV/EBIT	Comparaison secteur

Management	SC-MGT	/5	Tone analysis, insider	Pos/neg → pts
Moat	SC-MOA	/4	Gross margin, market s	Marge > seuil
Sentiment	SC-SEN	/3	News sentiment, analys	Sentiment → pts

BUY >= 28 | HOLD 18-27 | SELL < 18 | INSUFFISANT si < 3 sources valides (BR-002). Chaque point justifie par une regle (ex: 'PER < 15 → +2 pts').

3.2 AnalysisResult

ticker, company_name, sector, industry, score: Scoring|null, financials: FinancialData, sources: Source[] (>=1, scoring si >=3), management_tone, risks: RiskItem[] (<=20), pdf_path, dossier_path, generated_at, language: 'en'|'jp'|'bilingual'

4. Regles de gestion

- BR-001 — Pas d'invention — tout chiffre source et trace
- BR-002 —

5. Exigences non fonctionnelles

ID	Categorie	Exigence	Cible
NFR-00	Performance	Analyse complete	< 5 min
NFR-00	Performance	Affichage PDF	< 2 s
NFR-00	Disponibilite	API health	99% (cron 15min)
NFR-00	Securite	Auth write endpoint	X-API-Key (15 routes)
NFR-00	Fiabilite	Donnees manquantes	Pas d'invention
NFR-00	Observabilite	Logs structures	JSON + timestamp
NFR-00	Resilience	Rate limit APIs	Retry/backoff auto
NFR-00	Maintenabilit	Couverture tests	>= 60%
NFR-00	Securite	Secrets .env	Scan pre-commit

6. Architecture et deployment

6.1 Stack technique

Couche	Technologie
Backend	Python 3.11+ FastAPI (29 routes)
Frontend	React + Vite
PDF Engine	ReportLab 4.x
Scoring	Rules engine deterministe (40 pts, .py 140 lignes)
Sources	yfinance, Finnhub, EDGAR, Seeking Alpha, Tavily, Alph
Tunnel	Cloudflare Tunnel named (e92cfcfb)
Runtime	WSL2, venv, systemd user services
Tests	pytest, pytest-asyncio, Playwright E2E

6.2 Pipeline 9 etapes

- PIP-001 Profil → yfinance
- PIP-002 C

6.3 Securite

Couches de securite

- Auth: CED_CONTROL_KEY, X-API-Key sur 15 routes write/admin
- Rate limiter: 3 tiers (heavy 10/min, moderate 30/min, default 120/min)
- Bypass: localhost + same-origin (frontend sa.cedlabusa.net)
- CORS: sa.cedlabusa.net, localhost:5173, localhost:8780
- Cloudflare TLS: named tunnel, chiffrement bout en bout
- Secrets: .env uniquement, .gitignore, scan pre-commit

6.4 Sources de donnees

Source	Module	Rate limit	Timeout	Fallback
Yahoo Financ	yfinance	~5 req/s	15s	Finnhub
Finnhub	finnhub-pytho	60 req/min	10s	EDGAR
SEC EDGAR	edgartools	~10 req/s	20s	Aucun (primaire)
Seeking Alph	rapidapi_sa.p	Variable	25s	Tavily + AV
Tavily	tavily API	1000 req/mois	10s	DDG (degrade)
Alpha Vantag	alpha_vantage	25 req/jour	10s	Cache + Finnhub

7. Registre des risques

Medium **RSK-001: API yfinance depreciee**
Risque: yfinance non officiel — peut casser sans preavis
Mitigation: Finnhub + EDGAR en fallback. Health check 15 min.

Low **RSK-002: Double bookkeeping scorer**
Risque: Enrichissement EDGAR: financials vs fin desync
Mitigation: Fixe 3cf8f9b — sync systematique.

Low **RSK-003: Quick tunnel parasite**
Risque: cloudflared quick prend le port → instable
Mitigation: Auto-recovery detecte + kill quick.

Medium **RSK-004: Memoire WSL insuffisante**
Risque: >3 batch → OOM kill
Mitigation: Max 10 tickers. Clean memoire.

Low **RSK-005: Indentation PDF renderer**
Risque: Helper apres render → code absorbe
Mitigation: Fixe af2b3df — helpers AVANT.

High **RSK-006: Hallucination URLs PDF**
Risque: LLM invente URLs ou 404
Mitigation: BL-SA-003: URL checker.

Medium **RSK-007: Routes admin publiques**
Risque: 15 routes sans auth (avant fix)
Mitigation: Fixe 5110931: X-API-Key + rate limiter.

8. Glossaire

Ticker	Symbole boursier 1-5 caracteres uppercase (AAPL, NVDA, MSFT)
Deep-dive	Analyse narrative generee par LLM a partir des donnees collectees
Scoring	Note /40 sur 6 categories -> BUY(>=28)/HOLD(18-27)/SELL(<18)
INSUFFISANT	Verdict quand <3 sources valides — donnees insuffisantes
EDGAR	SEC Electronic Data Gathering — base de depots reglementaires US
Callout box	Encadre colore #2563EB dans le PDF pour information cle
Named tunnel	Tunnel Cloudflare permanent avec domaine stable (production)
Quick tunnel	Tunnel Cloudflare ephemere sans domaine (debug uniquement)
Bilingual	Mode EN+JP cote a cote — format modele.pdf original
CacheBusting	Cache-Control: no-cache sur fichiers statiques

- GA-001: Analyse NVDA complete < 5 min

- GA-002: